

College Attendance Management System

1. Introduction

The College Attendance Management System is a relational database system designed to manage and maintain student attendance records efficiently. The system stores information about students, faculty, departments, subjects, classes, and attendance.

In a traditional attendance system, maintaining records manually can be time-consuming and may lead to errors. This database-based system provides a structured way to record attendance, calculate attendance percentages, and generate useful reports.

2. Problem Statement

Managing attendance manually using registers or spreadsheets can result in:

- Loss or duplication of attendance records.
- Difficulty in calculating attendance percentages.
- Time-consuming report generation.
- Errors during data entry.
- Difficulty tracking attendance subject-wise.
- Difficulty identifying students with low attendance.

Therefore, a College Attendance Management System is required to securely store attendance data and provide quick access to accurate attendance information.

3. Objectives

The main objectives of the system are:

1. To maintain student information digitally.
2. To store department, faculty, and subject details.
3. To record daily attendance of students.
4. To calculate student attendance percentages.
5. To identify students having low attendance.
6. To reduce manual paperwork.
7. To maintain data consistency using database constraints.
8. To generate attendance reports efficiently.
9. To provide a normalized relational database structure.

4. ER Diagram

Main Entities

- Department
- Student
- Faculty
- Subject
- Class
- Attendance

5. Database Schema

5.1 Department Table

Attribute	Data Type	Constraint
Department_ID	INT	Primary Key
Department_Name	VARCHAR(100)	NOT NULL, UNIQUE

5.2 Student Table

Attribute	Data Type	Constraint
Student_ID	INT	Primary Key
Student_Name	VARCHAR(100)	NOT NULL
Email	VARCHAR(100)	UNIQUE
Phone	VARCHAR(15)	
Department_ID	INT	Foreign Key

5.3 Faculty Table

Attribute	Data Type	Constraint
Faculty_ID	INT	Primary Key
Faculty_Name	VARCHAR(100)	NOT NULL
Email	VARCHAR(100)	UNIQUE
Department_ID	INT	Foreign Key

5.4 Subject Table

Attribute	Data Type	Constraint
Subject_ID	INT	Primary Key
Subject_Name	VARCHAR(100)	NOT NULL
Faculty_ID	INT	Foreign Key
Department_ID	INT	Foreign Key

5.5 Attendance Table

Attribute	Data Type	Constraint
Attendance_ID	INT	Primary Key
Student_ID	INT	Foreign Key
Subject_ID	INT	Foreign Key
Attendance_Date	DATE	NOT NULL
Status	VARCHAR(10)	Present/Absent

6. Normalization

The database is designed using normalization principles to reduce redundancy and improve data consistency.

First Normal Form — 1NF

- Each column contains atomic values.
- There are no repeating groups.
- Each record has a unique primary key.

Second Normal Form — 2NF

- The database is already in 1NF.
- Non-key attributes depend completely on their primary key.

Third Normal Form — 3NF

- The database is in 2NF.
- There are no unnecessary transitive dependencies.
- Department information is stored separately instead of repeatedly storing department names in Student and Faculty records.

Thus, the proposed database follows 3NF.

7. Technologies Used

Technology	Purpose
MySQL	Relational database
SQL	Database queries
HTML	User interface
CSS	Interface design
JavaScript	Client-side functionality
PHP/Java/Python	Backend connectivity
MySQL Workbench	Database design and management

8.SQL Queries and Features Implemented

Create Database

```
CREATE DATABASE college_attendance;
```

```
USE college_attendance;
```

Create Department Table

```
CREATE TABLE department (
```

```
    department_id INT PRIMARY KEY AUTO_INCREMENT,
```

```
    department_name VARCHAR(100) NOT NULL UNIQUE
```

```
);
```

Create Student Table

```
CREATE TABLE student (
```

```
    student_id INT PRIMARY KEY AUTO_INCREMENT,
```

```
roll_no VARCHAR(20) NOT NULL UNIQUE,  
student_name VARCHAR(100) NOT NULL,  
email VARCHAR(100) UNIQUE,  
semester INT NOT NULL,  
department_id INT,  
FOREIGN KEY (department_id)  
REFERENCES department(department_id)  
ON DELETE SET NULL  
ON UPDATE CASCADE  
);
```

Create Faculty Table

```
CREATE TABLE faculty (  
faculty_id INT PRIMARY KEY AUTO_INCREMENT,  
faculty_name VARCHAR(100) NOT NULL,  
email VARCHAR(100) UNIQUE,  
department_id INT,  
FOREIGN KEY (department_id)  
REFERENCES department(department_id)  
ON DELETE SET NULL  
ON UPDATE CASCADE  
);
```

Create Subject Table

```
CREATE TABLE subject (  
subject_id INT PRIMARY KEY AUTO_INCREMENT,  
subject_code VARCHAR(20) NOT NULL UNIQUE,  
subject_name VARCHAR(100) NOT NULL,
```

```
faculty_id INT,  
  
FOREIGN KEY (faculty_id)  
  
REFERENCES faculty(faculty_id)  
  
ON DELETE SET NULL  
  
ON UPDATE CASCADE  
  
);
```

Create Attendance Table

```
CREATE TABLE attendance (  
  
attendance_id INT PRIMARY KEY AUTO_INCREMENT,  
  
student_id INT NOT NULL,  
  
subject_id INT NOT NULL,  
  
attendance_date DATE NOT NULL,  
  
status ENUM('Present', 'Absent') NOT NULL,  
  
  
FOREIGN KEY (student_id)  
  
REFERENCES student(student_id)  
  
ON DELETE CASCADE,  
  
  
FOREIGN KEY (subject_id)  
  
REFERENCES subject(subject_id)  
  
ON DELETE CASCADE,  
  
  
UNIQUE(student_id, subject_id, attendance_date)  
  
);
```

Insert Sample Data

INSERT INTO department (department_name)

VALUES

('Computer Science'),

('Information Technology'),

('Electronics Engineering');

INSERT INTO student

(roll_no, student_name, email, semester, department_id)

VALUES

('CS001', 'Rahul Patil', 'rahul@example.com', 5, 1),

('CS002', 'Amit Sharma', 'amit@example.com', 5, 1),

('IT001', 'Priya Singh', 'priya@example.com', 5, 2);

INSERT INTO faculty

(faculty_name, email, department_id)

VALUES

('Dr. Sharma', 'sharma@example.com', 1),

('Prof. Patil', 'patil@example.com', 1);

INSERT INTO subject

(subject_code, subject_name, faculty_id)

VALUES

('CS501', 'Database Management System', 1),

('CS502', 'Computer Networks', 2);

Record Attendance

INSERT INTO attendance

(student_id, subject_id, attendance_date, status)

VALUES

(1, 1, '2026-10-05', 'Present'),

(2, 1, '2026-10-05', 'Absent'),

(3, 1, '2026-10-05', 'Present');

Display All Students

SELECT * FROM student;

Display Attendance

SELECT

s.roll_no,

s.student_name,

sub.subject_name,

a.attendance_date,

a.status

FROM attendance a

JOIN student s

ON a.student_id = s.student_id

JOIN subject sub

ON a.subject_id = sub.subject_id;

Calculate Attendance Percentage

SELECT

s.roll_no,

```
s.student_name,  
  
sub.subject_name,  
  
COUNT(a.attendance_id) AS total_classes,  
  
SUM(a.status = 'Present') AS present_classes,  
  
ROUND(  
  
    SUM(a.status = 'Present') * 100.0 /  
  
    COUNT(a.attendance_id), 2  
  
    ) AS attendance_percentage  
  
FROM attendance a  
  
JOIN student s  
  
    ON a.student_id = s.student_id  
  
JOIN subject sub  
  
    ON a.subject_id = sub.subject_id  
  
GROUP BY  
  
    s.student_id,  
  
    sub.subject_id;
```

Find Students Below 75%

```
SELECT  
  
    s.roll_no,  
  
    s.student_name,  
  
    ROUND(  
  
        SUM(a.status = 'Present') * 100.0 /  
  
        COUNT(a.attendance_id), 2  
  
        ) AS attendance_percentage  
  
FROM attendance a
```

JOIN student s

ON a.student_id = s.student_id

GROUP BY s.student_id

HAVING attendance_percentage < 75;

9. Conclusion

The College Attendance Management System provides an efficient and organized solution for managing student attendance. By using a relational database, information about students, faculty, subjects, and attendance can be stored systematically.

The system reduces manual work, minimizes data-entry errors, provides quick access to attendance records, and makes it easier to calculate attendance percentages and identify students with low attendance.

Thus, the proposed system can improve the efficiency, accuracy, and reliability of college attendance management.

10. Future Scope

The system can be enhanced in the future by adding:

1. **Student and faculty login system**
2. **Admin dashboard**
3. **QR-code based attendance**
4. **Biometric attendance**
5. **Face-recognition attendance**
6. **Email/SMS notifications for low attendance**
7. **PDF attendance reports**
8. **Mobile application**
9. **Cloud database integration**
10. **Automatic monthly and semester reports**
11. **Role-based access control**
12. **Real-time attendance analytics and charts**

code:-

<html>

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
    <meta charset="UTF-8">
```

```
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <title>College Attendance Management System</title>
```

```
    <link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<!-- ===== HEADER ===== -->
```

```
<header class="header">
```

```
    <div class="logo">
```

```
        🎓 <span>College Attendance</span>
```

```
    </div>
```

```
    <div class="header-date" id="currentDate"></div>
```

```
</header>
```

<!-- ===== MAIN ===== -->

<main class="container">

<!-- HERO -->

<section class="hero">

<div>

<p class="small-title">COLLEGE MANAGEMENT</p>

<h1>Attendance Management System</h1>

<p>

Manage students, subjects and attendance
easily from one dashboard.

</p>

</div>

<div class="hero-icon">



</div>

</section>

```
<!-- ===== STATISTICS ===== -->
```

```
<section class="stats">
```

```
<div class="stat-card">
```

```
<div class="stat-icon purple">
```



```
</div>
```

```
<div>
```

```
<h2 id="totalStudents">0</h2>
```

```
<p>Total Students</p>
```

```
</div>
```

```
</div>
```

```
<div class="stat-card">
```

```
<div class="stat-icon blue">
```



```
</div>
```

```
<div>
```

```
<h2 id="totalSubjects">0</h2>
```

<p>Total Subjects</p>

</div>

</div>

<div class="stat-card">

<div class="stat-icon green">



</div>

<div>

<h2 id="todayPresent">0</h2>

<p>Present Today</p>

</div>

</div>

<div class="stat-card">

<div class="stat-icon red">



</div>

<div>

<h2 id="todayAbsent">0</h2>

<p>Absent Today</p>

</div>

</div>

</section>

<!-- ===== NAVIGATION ===== -->

<nav class="tabs">

<button class="tab active"

onclick="showSection('dashboardSection', this)">

 Dashboard

</button>

<button class="tab"

onclick="showSection('attendanceSection', this)">

 Mark Attendance

</button>

<button class="tab"

onclick="showSection('studentsSection', this)">

 Students

</button>

<button class="tab"
 onclick="showSection('subjectsSection', this)">

 Subjects

</button>

<button class="tab"
 onclick="showSection('recordsSection', this)">

 Records

</button>

</nav>

<!-- ===== DASHBOARD ===== -->

<section id="dashboardSection" class="section">

<div class="dashboard-grid">

<div class="card">

<h2>  Quick Actions</h2>

```
<div class="quick-actions">
```

```
<button
```

```
  onclick="openTab('attendanceSection')">
```

```
     Mark Attendance
```

```
</button>
```

```
<button
```

```
  onclick="openTab('studentsSection')">
```

```
     Add Student
```

```
</button>
```

```
<button
```

```
  onclick="openTab('subjectsSection')">
```

```
     Add Subject
```

```
</button>
```

```
<button
```

```
  onclick="openTab('recordsSection')">
```

```
     View Records
```

```
</button>
```

```
</div>
```

```
</div>
```

```
<div class="card">
```

```
<h2> Storage</h2>
```

```
<p class="storage-text">
```

Your data is stored locally in this
browser using localStorage.

```
</p>
```

```
<button
```

```
class="btn danger"
```

```
onclick="clearAllData()">
```

 Clear All Data

```
</button>
```

```
</div>
```

```
</div>
```

```
<div class="card">
```

```
<h2> Attendance Overview</h2>
```

```
<div id="dashboardOverview"
```

class="overview">

</div>

</div>

</section>

<!-- ===== ATTENDANCE ===== -->

<section id="attendanceSection"

class="section hidden">

<div class="card">

<div class="card-header">

<div>

<h2>  Mark Attendance</h2>

<p>Select subject and date.</p>

</div>

</div>

<div class="form-grid">

```
<div class="form-group">
```

```
<label>Subject</label>
```

```
<select id="attendanceSubject"
```

```
    onchange="loadAttendanceTable()"
```

```
    <option value="">
```

```
        Select Subject
```

```
    </option>
```

```
</select>
```

```
</div>
```

```
<div class="form-group">
```

```
<label>Date</label>
```

```
<input type="date"
```

```
    id="attendanceDate"
```

```
    onchange="loadAttendanceTable()"
```

```
</div>
```

</div>

<div class="attendance-buttons">

<button class="btn success"

onclick="markAll('Present')">

✓ Mark All Present

</button>

<button class="btn danger"

onclick="markAll('Absent')">

✗ Mark All Absent

</button>

</div>

<div class="table-container">

<table>

<thead>

<tr>

`<th>Roll No</th>`

`<th>Student Name</th>`

`<th>Department</th>`

`<th>Semester</th>`

`<th>Division</th>`

`<th>Status</th>`

`</tr>`

`</thead>`

`<tbody id="attendanceTable">`

`</tbody>`

`</table>`

`</div>`

`<button class="btn primary save-button"`

`onclick="saveAttendance()">`

` Save Attendance`

`</button>`

</div>

</section>

<!-- ===== STUDENTS ===== -->

<section id="studentsSection"

class="section hidden">

<div class="card">

<h2>🎓 Student Management</h2>

<div class="form-grid">

<div class="form-group">

<label>Roll Number *</label>

<input type="text"

id="rollNo"

placeholder="CSE001">

</div>

```
<div class="form-group">
```

```
<label>Student Name *</label>
```

```
<input type="text"
```

```
  id="studentName"
```

```
  placeholder="Enter student name">
```

```
</div>
```

```
<div class="form-group">
```

```
<label>Department *</label>
```

```
<input type="text"
```

```
  id="department"
```

```
  placeholder="Computer Engineering">
```

```
</div>
```

```
<div class="form-group">
```

```
<label>Semester *</label>
```

<select id="semester">

<option value="1">1</option>

<option value="2">2</option>

<option value="3">3</option>

<option value="4">4</option>

<option value="5">5</option>

<option value="6">6</option>

<option value="7">7</option>

<option value="8">8</option>

</select>

</div>

<div class="form-group">

<label>Division</label>

<input type="text"

id="division"

placeholder="A">

</div>

</div>

<button class="btn primary"
 onclick="addStudent()">

✚ Add Student

</button>

</div>

<div class="card">

<div class="card-header">

<h2>Student List</h2>

<input

 class="search"

 id="studentSearch"

 type="text"

 placeholder="🔍 Search student..."

 oninput="displayStudents()">

</div>

<div class="table-container">

<table>

<thead>

<tr>

<th>#</th>

<th>Roll No</th>

<th>Name</th>

<th>Department</th>

<th>Semester</th>

<th>Division</th>

<th>Action</th>

</tr>

</thead>

<tbody id="studentsTable">

</tbody>

</table>

</div>

</div>

</section>

<!-- ===== SUBJECTS ===== -->

<section id="subjectsSection"

class="section hidden">

<div class="card">

<h2> Subject Management</h2>

<div class="form-grid">

<div class="form-group">

<label>Subject Code *</label>

<input type="text"

id="subjectCode"

placeholder="CS501">

</div>

<div class="form-group">

<label>Subject Name *</label>

<input type="text"

id="subjectName"

placeholder="Database Management System">

</div>

</div>

<button class="btn primary"

onclick="addSubject()">

✚ Add Subject

</button>

</div>

```
<div class="card">
```

```
<h2>Subject List</h2>
```

```
<div class="table-container">
```

```
<table>
```

```
<thead>
```

```
<tr>
```

```
<th>#</th>
```

```
<th>Code</th>
```

```
<th>Subject Name</th>
```

```
<th>Action</th>
```

```
</tr>
```

```
</thead>
```

```
<tbody id="subjectsTable">
```

```
</tbody>
```

```
</table>
```

</div>

</div>

</section>

<!-- ===== RECORDS ===== -->

<section id="recordsSection"
class="section hidden">

<div class="card">

<h2> Attendance Records</h2>

<div class="filter-grid">

<div class="form-group">

<label>Subject</label>

<select id="recordSubject"
onchange="displayRecords()">

`<option value="">`

`All Subjects`

`</option>`

`</select>`

`</div>`

`<div class="form-group">`

`<label>Date</label>`

`<input type="date"`

`id="recordDate"`

`onchange="displayRecords()">`

`</div>`

`<div class="form-group">`

`<label>Search</label>`

`<input type="text"`

`id="recordSearch"`

placeholder="Roll No / Name"

oninput="displayRecords()">

</div>

</div>

<div class="table-container">

<table>

<thead>

<tr>

<th>Date</th>

<th>Roll No</th>

<th>Student</th>

<th>Subject</th>

<th>Status</th>

<th>Action</th>

</tr>

</thead>

<tbody id="recordsTable">

</tbody>

</table>

</div>

</div>

<!-- PERCENTAGE -->

<div class="card">

<h2> Attendance Percentage</h2>

<div class="table-container">

<table>

<thead>

<tr>

<th>Roll No</th>

<th>Student</th>

<th>Total Classes</th>

<th>Present</th>

<th>Absent</th>

<th>Percentage</th>

</tr>

</thead>

<tbody id="percentageTable">

</tbody>

</table>

</div>

</div>

</section>

</main>

<!-- ===== FOOTER ===== -->

<footer>

<h3>🎓 College Attendance Management System</h3>

<p>

HTML • CSS • JavaScript • localStorage

</p>

<p>

No Login Required

</p>

</footer>

<script src="script.js"></script>

</body>

</html>

#style.css

* {

margin: 0;

padding: 0;

box-sizing: border-box;

}

:root {

--primary: #6d28d9;

--primary-dark: #4c1d95;

--purple-light: #ede9fe;

```
--blue: #2563eb;

--green: #16a34a;

--red: #dc2626;

--orange: #ea580c;


--background: #f5f7fb;

--white: #ffffff;

--text: #1f2937;

--muted: #6b7280;

--border: #e5e7eb;

}


body {

    font-family: Arial, Helvetica, sans-serif;


    background: var(--background);


    color: var(--text);


    min-height: 100vh;

}


/* ===== HEADER ===== */


.header {
```

background:

linear-gradient(

135deg,

#24104f,

#6d28d9

);

color: white;

padding: 18px 5%;

display: flex;

justify-content: space-between;

align-items: center;

position: sticky;

top: 0;

z-index: 100;

box-shadow:

0 5px 20px rgba(0,0,0,.15);

```
}
```

```
.logo {
```

```
    font-size: 22px;
```

```
    font-weight: bold;
```

```
}
```

```
.header-date {
```

```
    font-size: 14px;
```

```
    opacity: .9;
```

```
}
```

```
/* ===== CONTAINER ===== */
```

```
.container {
```

```
    width: 92%;
```

```
    max-width: 1400px;
```

```
    margin: 30px auto;
```

```
}
```

```
/* ===== HERO ===== */
```

```
.hero {
```

```
background:
```

```
  linear-gradient(
```

```
    135deg,
```

```
    #4c1d95,
```

```
    #7c3aed
```

```
  );
```

```
color: white;
```

```
border-radius: 20px;
```

```
padding: 35px;
```

```
display: flex;
```

```
align-items: center;
```

```
justify-content: space-between;
```

margin-bottom: 25px;

box-shadow:

0 10px 30px

rgba(76,29,149,.25);

}

.small-title {

font-size: 12px;

letter-spacing: 2px;

opacity: .8;

margin-bottom: 8px;

}

.hero h1 {

font-size: 32px;

margin-bottom: 10px;

}

.hero p {

opacity: .9;

line-height: 1.6;

}

.hero-icon {

font-size: 80px;

opacity: .8;

}

/* ===== STATS ===== */

.stats {

display: grid;

grid-template-columns:

repeat(4, 1fr);

gap: 20px;

margin-bottom: 25px;

```
}
```

```
.stat-card {
```

```
background: white;
```

```
border-radius: 16px;
```

```
padding: 22px;
```

```
display: flex;
```

```
align-items: center;
```

```
gap: 15px;
```

```
box-shadow:
```

```
0 5px 20px
```

```
rgba(0,0,0,.06);
```

```
}
```

```
.stat-icon {
```

```
width: 55px;
```

```
height: 55px;
```

border-radius: 14px;

display: flex;

justify-content: center;

align-items: center;

font-size: 25px;

}

.stat-icon.purple {

background: #ede9fe;

}

.stat-icon.blue {

background: #dbeafe;

}

.stat-icon.green {

background: #dcfce7;

}

```
.stat-icon.red {  
  
    background: #fee2e2;  
  
}
```

```
.stat-card h2 {  
  
    font-size: 28px;  
  
    margin-bottom: 4px;  
  
}
```

```
.stat-card p {  
  
    color: var(--muted);  
  
    font-size: 14px;  
  
}
```

```
/* ===== TABS ===== */
```

```
.tabs {  
  
    display: flex;
```

```
flex-wrap: wrap;

gap: 10px;

margin-bottom: 20px;
}

.tab {

background: white;

color: #374151;

border: 1px solid var(--border);

padding: 13px 20px;

border-radius: 10px;

cursor: pointer;

font-weight: bold;

transition: .2s;
}
```

```
.tab:hover {  
  
    transform: translateY(-2px);  
  
}
```

```
.tab.active {  
  
    background: var(--primary);  
  
    color: white;  
  
    border-color: var(--primary);  
  
}
```

```
/* ===== SECTION ===== */
```

```
.hidden {  
  
    display: none !important;  
  
}
```

```
/* ===== CARD ===== */
```

```
.card {  
  
    background: white;  
  
    border-radius: 16px;  
  
    padding: 25px;  
  
    margin-bottom: 25px;  
  
    box-shadow:  
        0 5px 20px  
        rgba(0,0,0,.06);  
}
```

```
.card h2 {  
  
    color: #312e81;  
  
    margin-bottom: 20px;  
}
```

```
.card-header {  
  
    display: flex;
```

```
justify-content: space-between;

align-items: center;

gap: 20px;

margin-bottom: 20px;
}

/* ===== DASHBOARD ===== */

.dashboard-grid {

display: grid;

grid-template-columns:

    2fr 1fr;

gap: 20px;
}

.quick-actions {

display: grid;
```

grid-template-columns:

repeat(2, 1fr);

gap: 15px;

}

.quick-actions button {

background: #f5f3ff;

color: #4c1d95;

border: 1px solid #ddd6fe;

padding: 20px;

border-radius: 12px;

cursor: pointer;

font-weight: bold;

font-size: 15px;

transition: .2s;

}

```
.quick-actions button:hover {
```

```
background: #ede9fe;
```

```
transform: translateY(-2px);
```

```
}
```

```
.storage-text {
```

```
color: var(--muted);
```

```
line-height: 1.6;
```

```
margin-bottom: 20px;
```

```
}
```

```
.overview {
```

```
display: grid;
```

```
grid-template-columns:
```

```
repeat(3, 1fr);
```

```
gap: 15px;
```

```
}
```

```
.overview-box {  
  
    background: #f9fafb;  
  
    border: 1px solid var(--border);  
  
    padding: 20px;  
  
    border-radius: 12px;  
  
    text-align: center;  
}
```

```
.overview-box h3 {  
  
    font-size: 25px;  
  
    margin-bottom: 5px;  
}
```

```
.overview-box p {  
  
    color: var(--muted);  
}
```

```
/* ===== FORMS ===== */
```

```
.form-grid {
```

```
display: grid;
```

```
grid-template-columns:
```

```
repeat(2, 1fr);
```

```
gap: 18px;
```

```
margin-bottom: 20px;
```

```
}
```

```
.filter-grid {
```

```
display: grid;
```

```
grid-template-columns:
```

```
repeat(3, 1fr);
```

```
gap: 18px;
```

```
margin-bottom: 25px;
```

```
}
```

```
.form-group {
```

```
display: flex;
```

```
flex-direction: column;
```

```
gap: 7px;
```

```
}
```

```
.form-group label {
```

```
font-size: 14px;
```

```
font-weight: bold;
```

```
}
```

```
input,
```

```
select {
```

```
width: 100%;
```

```
padding: 12px 14px;
```

```
border: 1px solid #d1d5db;
```

border-radius: 9px;

font-size: 15px;

background: white;

outline: none;

}

input:focus,

select:focus {

border-color: var(--primary);

box-shadow:

0 0 0 3px

rgba(109,40,217,.1);

}

.search {

max-width: 280px;

}

/* ===== BUTTONS ===== */

```
.btn {
```

```
border: none;
```

```
padding: 12px 18px;
```

```
border-radius: 9px;
```

```
cursor: pointer;
```

```
font-weight: bold;
```

```
transition: .2s;
```

```
}
```

```
.btn:hover {
```

```
transform: translateY(-1px);
```

```
opacity: .9;
```

```
}
```

```
.primary {
```

```
background: var(--primary);
```

```
    color: white;
}
```

```
.success {
```

```
    background: var(--green);
```

```
    color: white;
}
```

```
.danger {
```

```
    background: var(--red);
```

```
    color: white;
}
```

```
.warning {
```

```
    background: var(--orange);
```

```
    color: white;
}
```

```
/* ===== ATTENDANCE ===== */
```

```
.attendance-buttons {
```

```
    display: flex;
```

```
    gap: 10px;
```

```
    margin-bottom: 20px;
```

```
    flex-wrap: wrap;
```

```
}
```

```
.save-button {
```

```
    width: 100%;
```

```
    margin-top: 20px;
```

```
    font-size: 16px;
```

```
}
```

```
/* ===== TABLE ===== */
```

```
.table-container {
```

```
    overflow-x: auto;
}
```

```
table {
```

```
    width: 100%;
```

```
    border-collapse: collapse;
```

```
    min-width: 750px;
```

```
}
```

```
th {
```

```
    background: #f3f4f6;
```

```
    color: #374151;
```

```
    text-align: left;
```

```
    padding: 14px;
```

```
    font-size: 14px;
```

```
}
```

```
td {  
  
    padding: 13px 14px;  
  
    border-bottom:  
        1px solid var(--border);  
}
```

```
tr:hover {  
  
    background: #fafafa;  
}
```

```
/* ===== STATUS ===== */
```

```
.status {  
  
    border: none;  
  
    padding: 8px 14px;  
  
    border-radius: 20px;  
  
    font-weight: bold;
```

```
    cursor: pointer;

}
```

```
.status.present {
```

```
    background: #dcfce7;
```

```
    color: #166534;
```

```
}
```

```
.status.absent {
```

```
    background: #fee2e2;
```

```
    color: #991b1b;
```

```
}
```

```
/* ===== PERCENTAGE ===== */
```

```
.percentage {
```

```
    font-weight: bold;
```

```
    padding: 7px 12px;
```

border-radius: 20px;

display: inline-block;

}

.percentage.good {

background: #dcfce7;

color: #166534;

}

.percentage.warning {

background: #fef3c7;

color: #92400e;

}

.percentage.low {

background: #fee2e2;

color: #991b1b;

}

```
/* ===== FOOTER ===== */
```

```
footer {
```

```
background: #111827;
```

```
color: #d1d5db;
```

```
text-align: center;
```

```
padding: 30px;
```

```
margin-top: 40px;
```

```
line-height: 1.8;
```

```
}
```

```
footer h3 {
```

```
color: white;
```

```
margin-bottom: 8px;
```

```
}
```

```
/* ===== RESPONSIVE ===== */
```

```
@media(max-width: 1000px) {
```

```
  .stats {
```

```
    grid-template-columns:
```

```
      repeat(2, 1fr);
```

```
  }
```

```
  .dashboard-grid {
```

```
    grid-template-columns: 1fr;
```

```
  }
```

```
}
```

```
@media(max-width: 700px) {
```

```
  .header {
```

```
    flex-direction: column;
```

```
    gap: 10px;
```

```
text-align: center;  
}
```

```
.hero {
```

```
padding: 25px;  
  
}
```

```
.hero h1 {
```

```
font-size: 25px;  
}
```

```
.hero-icon {
```

```
font-size: 50px;  
}
```

```
.form-grid,  
.filter-grid {
```

```
grid-template-columns: 1fr;  
}
```

```
.overview {
```

```
    grid-template-columns: 1fr;  
}
```

```
.quick-actions {
```

```
    grid-template-columns: 1fr;  
}
```

```
.card-header {
```

```
    flex-direction: column;  
  
    align-items: stretch;  
}
```

```
.search {
```

```
    max-width: none;  
}
```

```
}
```

```
@media(max-width: 500px) {
```

```
.container {  
  
    width: 95%;  
  
    margin-top: 15px;  
}
```

```
.stats {  
  
    grid-template-columns: 1fr;  
}
```

```
.hero {  
  
    flex-direction: column;  
  
    align-items: flex-start;  
  
    gap: 15px;  
}
```

```
.tabs {  
  
    flex-direction: column;  
}
```

```

.tab {

    width: 100%;

}

}

Java Script.js

/* =====

COLLEGE ATTENDANCE MANAGEMENT SYSTEM

HTML + CSS + JavaScript + localStorage

===== */


/* =====

LOCAL STORAGE

===== */


const STUDENTS_KEY = "college_students";

const SUBJECTS_KEY = "college_subjects";

const ATTENDANCE_KEY = "college_attendance";


let students =

    JSON.parse(

        localStorage.getItem(STUDENTS_KEY)

    ) || [];

```

```
let subjects =  
  
    JSON.parse(  
  
        localStorage.getItem(SUBJECTS_KEY)  
  
    ) || [];
```

```
let attendance =  
  
    JSON.parse(  
  
        localStorage.getItem(ATTENDANCE_KEY)  
  
    ) || [];
```

```
/* =====  
  
    INITIALIZATION  
  
    ===== */
```

```
document.addEventListener(  
  
    "DOMContentLoaded",  
  
    function () {  
  
  
        setCurrentDate();  
  
  
        setDefaultDates();
```

```
createDemoDataIfEmpty();

refreshEverything();

}

);

/* =====

DEMO DATA

===== */

function createDemoDataIfEmpty() {

    if (students.length === 0) {

        students = [

            {

                id: generateId(),

                rollNo: "CSE001",

                name: "Rahul Patil",

                department:
```

"Computer Engineering",

semester: "5",

division: "A"

},

{

id: generateId(),

rollNo: "CSE002",

name: "Priya Sharma",

department:

"Computer Engineering",

semester: "5",

division: "A"

},

{

id: generateId(),

rollNo: "CSE003",

name: "Amit Joshi",

department:

"Computer Engineering",

semester: "5",

division: "A"

}

];

saveStudents();

}

if (subjects.length === 0) {

subjects = [

{

id: generateId(),

code: "CS501",

name: "Database Management System"

},

{

id: generateId(),

code: "CS502",

name: "Computer Networks"

},

{

id: generateId(),

code: "CS503",

name: "Software Engineering"

}

];

saveSubjects();

}

}

```
/* =====  
  
ID GENERATOR  
  
===== */  
  
function generateId() {  
  
    return Date.now() +  
  
        Math.floor(  
  
            Math.random() * 10000  
  
        );  
}
```

```
/* =====  
  
LOCAL STORAGE SAVE  
  
===== */  
  
function saveStudents() {  
  
    localStorage.setItem(  
  
        STUDENTS_KEY,  
  
        JSON.stringify(students)  
  
    );  
}
```

```
function saveSubjects() {  
  
    localStorage.setItem(  
        SUBJECTS_KEY,  
        JSON.stringify(subjects)  
    );  
}
```

```
function saveAttendance() {  
  
    localStorage.setItem(  
        ATTENDANCE_KEY,  
        JSON.stringify(attendance)  
    );  
}
```

```
/* =====  
  
    REFRESH EVERYTHING  
  
    ===== */
```

```
function refreshEverything() {  
  
    displayStudents();
```

displaySubjects();

populateSubjectDropdowns();

loadAttendanceTable();

displayRecords();

displayPercentages();

updateDashboard();

}

/* =====

CURRENT DATE

===== */

function setCurrentDate() {

const today = new Date();

document.getElementById(

"currentDate"

```

).textContent =

    today.toLocaleDateString(

        "en-IN",

        {

            weekday: "long",

            year: "numeric",

            month: "long",

            day: "numeric"

        }

    );
}

```

```

/* =====

DEFAULT DATES

===== */

```

```

function getToday() {

    const now = new Date();

    const year =

        now.getFullYear();

    const month =

        String(

```

```
        now.getMonth() + 1

    ).padStart(2, "0");

const day =

    String(

        now.getDate()

    ).padStart(2, "0");

return `${year}-${month}-${day}`;

}
```

```
function setDefaultDates() {

    const today = getToday();

    document.getElementById(

        "attendanceDate"

    ).value = today;

    document.getElementById(

        "recordDate"

    ).value = "";

}
```

```
/* =====
```

NAVIGATION

```
===== */
```

```
function showSection(
```

```
    sectionId,
```

```
    button
```

```
) {
```

```
    document
```

```
        .querySelectorAll(".section")
```

```
        .forEach(
```

```
            section => {
```

```
                section.classList.add(
```

```
                    "hidden"
```

```
                );
```

```
            }
```

```
        );
```

```
const section =
```

```
    document.getElementById(
```

```
        sectionId
```

```
    );
```

```
section.classList.remove(
```

"hidden"

);

document

.querySelectorAll(".tab")

.forEach(

tab => {

tab.classList.remove(

"active"

);

}

);

button.classList.add(

"active"

);

if (

sectionId ===

"attendanceSection"

) {

loadAttendanceTable();

```
}
```

```
if (
```

```
    sectionId ===
```

```
    "recordsSection"
```

```
) {
```

```
    displayRecords();
```

```
    displayPercentages();
```

```
}
```

```
}
```

```
function openTab(sectionId) {
```

```
    document
```

```
        .querySelectorAll(".section")
```

```
        .forEach(
```

```
            section => {
```

```
                section.classList.add(
```

```
                    "hidden"
```

```
);  
  
}  
  
);
```

document

```
.getElementById(sectionId)  
.classList.remove(  
    "hidden"  
);
```

document

```
.querySelectorAll(".tab")  
.forEach(  
    tab => {  
        tab.classList.remove(  
            "active"  
        );  
    }  
);
```

const tabs =

```
document.querySelectorAll(  
    ".tab"
```

);

tabs.forEach(tab => {

if (

tab.textContent

.toLowerCase()

.includes(

sectionId

.replace(

"Section",

""

)

.toLowerCase()

)

){

tab.classList.add(

"active"

);

}

});

```
}
```

```
/* =====
```

```
DASHBOARD
```

```
===== */
```

```
function updateDashboard() {
```

```
    document.getElementById(
```

```
        "totalStudents"
```

```
    ).textContent =
```

```
        students.length;
```

```
    document.getElementById(
```

```
        "totalSubjects"
```

```
    ).textContent =
```

```
        subjects.length;
```

```
    const today =
```

```
        getToday();
```

```
    const todayRecords =
```

```
attendance.filter(  
  record =>  
    record.date === today  
);
```

```
const present =  
  todayRecords.filter(  
    record =>  
      record.status ===  
        "Present"  
  ).length;
```

```
const absent =  
  todayRecords.filter(  
    record =>  
      record.status ===  
        "Absent"  
  ).length;
```

```
document.getElementById(  
  "todayPresent"  
).textContent =  
  present;
```

```
document.getElementById(  
    "todayAbsent"  
).textContent =  
    absent;
```

```
const overview =  
    document.getElementById(  
        "dashboardOverview"  
    );
```

```
const total =  
    present + absent;
```

```
const percentage =  
    total > 0  
  
    ? (  
        present /  
        total *  
        100  
    ).toFixed(1)
```

: 0;

overview.innerHTML = `

<div class="overview-box">

<h3>\${students.length}</h3>

<p>Students Registered</p>

</div>

<div class="overview-box">

<h3>\${subjects.length}</h3>

<p>Subjects Added</p>

</div>

<div class="overview-box">

```
<h3>${percentage}%</h3>
```

```
<p>Today's Attendance</p>
```

```
</div>
```

```
`;  
`;
```

```
}
```

```
/* =====
```

```
STUDENTS
```

```
===== */
```

```
function addStudent() {
```

```
    const rollNo =
```

```
        document.getElementById(
```

```
            "rollNo"
```

```
        ).value.trim();
```

```
    const name =
```

```
        document.getElementById(
```

```
            "studentName"
```

```
).value.trim();
```

```
const department =
```

```
    document.getElementById(
```

```
        "department"
```

```
).value.trim();
```

```
const semester =
```

```
    document.getElementById(
```

```
        "semester"
```

```
).value;
```

```
const division =
```

```
    document.getElementById(
```

```
        "division"
```

```
).value.trim();
```

```
if (
```

```
    !rollNo ||
```

```
    !name ||
```

```
    !department
```

```
) {
```

```
    alert(  
        "Please fill all required fields."  
    );  
  
    return;  
}  
  
const duplicate =  
    students.some(  
        student =>  
            student.rollNo  
                .toLowerCase() ===  
                rollNo.toLowerCase()  
    );  
  
if (duplicate) {  
  
    alert(  
        "This roll number already exists."  
    );  
  
    return;  
}
```

```
students.push({

    id: generateId(),

    rollNo: rollNo,

    name: name,

    department: department,

    semester: semester,

    division: division

});

saveStudents();

document.getElementById(

    "rollNo"

).value = "";
```

```
document.getElementById(  
    "studentName"  
).value = "";
```

```
document.getElementById(  
    "department"  
).value = "";
```

```
document.getElementById(  
    "division"  
).value = "";
```

```
refreshEverything();
```

```
alert(  
    "Student added successfully! ✓"  
);
```

```
}
```

```
/* =====
```

```
DISPLAY STUDENTS
```

```
===== */
```

```
function displayStudents() {
```

```
    const table =
```

```
        document.getElementById(
```

```
            "studentsTable"
```

```
        );
```

```
    const search =
```

```
        document.getElementById(
```

```
            "studentSearch"
```

```
        ).value
```

```
        .toLowerCase()
```

```
        .trim() || "";
```

```
    const filtered =
```

```
        students.filter(
```

```
            student =>
```

```
                student.rollNo
```

```
                    .toLowerCase()
```

`.includes(search)`

`||`

`student.name`

`.toLowerCase()`

`.includes(search)`

`||`

`student.department`

`.toLowerCase()`

`.includes(search)`

`);`

`table.innerHTML = "";`

`if (`

`filtered.length === 0`

`) {`

`table.innerHTML = ``

`<tr>`

```
<td colspan="7"
```

```
style="text-align:center">
```

```
No students found.
```

```
</td>
```

```
</tr>
```

```
`;
```

```
return;
```

```
}
```

```
filtered.forEach(
```

```
(student, index) => {
```

```
table.innerHTML += `
```

```
<tr>
```

```
<td>
```

```
  ${index + 1}
```

```
</td>
```

<td>

 \${escapeHTML(
 student.rollNo
)}

</td>

<td>

 \${escapeHTML(
 student.name
)}

</td>

<td>

 \${escapeHTML(
 student.department
)}

</td>

<td>

 \${student.semester}

</td>

<td>

```
    ${escapeHTML(
        student.division || "-"
    )}
```

```
</td>
```

```
<td>
```

```
<button
    class="btn warning"
    onclick="editStudent('${student.id}')">
```

 Edit

```
</button>
```

```
<button
    class="btn danger"
    onclick="deleteStudent('${student.id}')">
```

 Delete

```
</button>
```

```
</td>
```

```
</tr>
```

```
`;  
`
```

```
}  
`
```

```
);  
`
```

```
}  
`
```

```
/* =====  
`
```

```
EDIT STUDENT  
`
```

```
===== */  
`
```

```
function editStudent(id) {  
`
```

```
const student =  
`
```

```
students.find(  
`
```

```
item =>  
`
```

```
String(item.id) ===  
`
```

```
String(id)  
`
```

```
);  
`
```

```
if (!student) return;  
`
```

```
const newName =  
  prompt(  
    "Enter student name:",  
    student.name  
  );
```

```
if (  
  newName === null ||  
  !newName.trim()  
) {  
  
  return;  
}
```

```
const newDepartment =  
  prompt(  
    "Enter department:",  
    student.department  
  );
```

```
if (  
  newDepartment === null ||  
  !newDepartment.trim()
```

```
) {  
  
    return;  
}  
  
student.name =  
    newName.trim();  
  
student.department =  
    newDepartment.trim();  
  
saveStudents();  
  
refreshEverything();  
  
alert(  
    "Student updated successfully! ✓"  
);  
  
}
```

```
/* =====
```

```
DELETE STUDENT
```

```
===== */
```

```
function deleteStudent(id) {
```

```
  const student =
```

```
    students.find(
```

```
      item =>
```

```
        String(item.id) ===
```

```
        String(id)
```

```
    );
```

```
  if (!student) return;
```

```
  const confirmDelete =
```

```
    confirm(
```

```
      `Delete ${student.name}?`
```

```
    );
```

```
  if (!confirmDelete) return;
```

students =

students.filter(

item =>

String(item.id) !==

String(id)

);

attendance =

attendance.filter(

record =>

String(record.studentId) !==

String(id)

);

saveStudents();

saveAttendance();

refreshEverything();

alert(

"Student deleted successfully."

);

```
}
```

```
/* =====
```

```
SUBJECTS
```

```
===== */
```

```
function addSubject() {
```

```
    const code =
```

```
        document.getElementById(
```

```
            "subjectCode"
```

```
        ).value.trim();
```

```
    const name =
```

```
        document.getElementById(
```

```
            "subjectName"
```

```
        ).value.trim();
```

```
    if (!code || !name) {
```

```
        alert(
```

```
            "Please enter subject code and name."
```

);

return;

}

const duplicate =

subjects.some(

subject =>

subject.code

.toLowerCase() ===

code.toLowerCase()

);

if (duplicate) {

alert(

"This subject code already exists."

);

return;

}

subjects.push({

id: generateId(),

code: code,

name: name

});

saveSubjects();

document.getElementById(

"subjectCode"

).value = "";

document.getElementById(

"subjectName"

).value = "";

refreshEverything();

```

alert(

    "Subject added successfully! ✓"

);

}

/* =====

DISPLAY SUBJECTS

===== */

function displaySubjects() {

    const table =

        document.getElementById(

            "subjectsTable"

        );

    table.innerHTML = "";

    if (

        subjects.length === 0

    ) {

```

```
table.innerHTML = `
```

```
<tr>
```

```
<td colspan="4"
```

```
style="text-align:center">
```

```
No subjects available.
```

```
</td>
```

```
</tr>
```

```
`;
```

```
return;
```

```
}
```

```
subjects.forEach(
```

```
(subject, index) => {
```

```
table.innerHTML += `
```

```
<tr>
```

<td>

`${index + 1}`

</td>

<td>

`${escapeHTML(`

`subject.code`

`)}`

</td>

<td>

`${escapeHTML(`

`subject.name`

`)}`

</td>

<td>

<button

`class="btn danger"`

`onclick="deleteSubject('${subject.id}')">`

 Delete

</button>

</td>

</tr>

`;

}

);

}

/* =====

DELETE SUBJECT

===== */

function deleteSubject(id) {

const subject =

subjects.find(

item =>

String(item.id) ===

String(id)

);

```
if (!subject) return;
```

```
if (  
    !confirm(  
        `Delete ${subject.name}?`  
    )  
) {  
  
    return;  
}
```

```
subjects =  
    subjects.filter(  
        item =>  
            String(item.id) !==  
            String(id)  
    );
```

```
attendance =  
    attendance.filter(  
        record =>
```

```
        String(record.subjectId) !==  
        String(id)  
    );  
  
    saveSubjects();  
  
    saveAttendance();  
  
    refreshEverything();  
  
    alert(  
        "Subject deleted successfully."  
    );  
  
}  
  
/* =====  
  
SUBJECT DROPDOWNS  
  
===== */  
  
function populateSubjectDropdowns() {  
  
    const attendanceSelect =
```

```
document.getElementById(  
    "attendanceSubject"  
);
```

```
const recordSelect =  
    document.getElementById(  
        "recordSubject"  
    );
```

```
const oldAttendanceValue =  
    attendanceSelect.value;
```

```
const oldRecordValue =  
    recordSelect.value;
```

```
attendanceSelect.innerHTML = `
```

```
<option value="">
```

```
    Select Subject
```

```
</option>
```

```
`;  
`;
```

```
recordSelect.innerHTML = `
```

```
  <option value="">
```

```
    All Subjects
```

```
  </option>
```

```
`;
```

```
subjects.forEach(
```

```
  subject => {
```

```
    attendanceSelect.innerHTML += `
```

```
      <option value="${subject.id}">
```

```
        ${escapeHTML(
```

```
          subject.code
```

```
        )}
```

```
      -
```

```
      ${escapeHTML(
```

```
        subject.name
```

```
      )}
```

</option>

`;

recordSelect.innerHTML += `

<option value="\${subject.id}">

\${escapeHTML(

subject.code

)}

-

\${escapeHTML(

subject.name

)}

</option>

`;

}

);

if (

```
subjects.some(  
  s =>  
    String(s.id) ===  
      oldAttendanceValue  
)  
{  
  
  attendanceSelect.value =  
    oldAttendanceValue;  
  
}
```

```
if (  
  subjects.some(  
    s =>  
      String(s.id) ===  
        oldRecordValue  
  )  
{  
  
  recordSelect.value =  
    oldRecordValue;  
  
}
```

```
}
```

```
/* =====
```

```
ATTENDANCE TABLE
```

```
===== */
```

```
function loadAttendanceTable() {
```

```
    const table =
```

```
        document.getElementById(
```

```
            "attendanceTable"
```

```
        );
```

```
    const subjectId =
```

```
        document.getElementById(
```

```
            "attendanceSubject"
```

```
        ).value;
```

```
    const date =
```

```
        document.getElementById(
```

```
            "attendanceDate"
```

```
        ).value;
```

```
table.innerHTML = "";
```

```
if (!subjectId || !date) {
```

```
    table.innerHTML = `
```

```
        <tr>
```

```
            <td colspan="6"
```

```
                style="text-align:center">
```

```
                Select a subject and date
```

```
                to mark attendance.
```

```
            </td>
```

```
        </tr>
```

```
    `;
```

```
    return;
```

```
}
```

```
if (  
    students.length === 0  
) {  
  
    table.innerHTML = `  
  
        <tr>  
  
            <td colspan="6"  
                style="text-align:center">  
  
                    Add students first.  
  
            </td>  
  
        </tr>  
  
    `;  
  
    return;  
}  
  
students.forEach(  
    student => {
```

const existing =

attendance.find(

record =>

String(

record.studentId

) ===

String(student.id)

&&

String(

record.subjectId

) ===

String(subjectId)

&&

record.date ===

date

);

const status =

existing

? existing.status

```
: "Present";
```

```
const isPresent =
```

```
    status === "Present";
```

```
table.innerHTML += `
```

```
<tr>
```

```
<td>
```

```
<strong>
```

```
    ${escapeHTML(  
        student.rollNo  
    )}
```

```
</strong>
```

```
</td>
```

```
<td>
```

```
    ${escapeHTML(  
        student.name  
    )}
```

```
</td>
```

```
<td>
```

```
        ${escapeHTML(
            student.department
        )}
    </td>
```

```
<td>
    ${student.semester}
</td>
```

```
<td>
    ${escapeHTML(
        student.division || "-"
    )}
</td>
```

```
<td>
```

```
<button
```

```
    class="status
```

```
    ${isPresent
```

```
        ? "present"
```

```
        : "absent"}"
```

```
    data-student="${student.id}"
```

data-status="\${status}"

onclick="toggleStatus(this)">

\${isPresent

? "✓ Present"

: "✗ Absent"}

</button>

</td>

</tr>

`;

}

);

}

/* =====

TOGGLE ATTENDANCE STATUS

===== */

```
function toggleStatus(button) {
```

```
    const current =
```

```
        button.dataset.status;
```

```
    if (
```

```
        current === "Present"
```

```
    ) {
```

```
        button.dataset.status =
```

```
            "Absent";
```

```
        button.textContent =
```

```
            "× Absent";
```

```
        button.classList.remove(
```

```
            "present"
```

```
        );
```

```
        button.classList.add(
```

```
            "absent"
```

```
        );
```

```
    } else {
```

```
button.dataset.status =  
    "Present";  
  
button.textContent =  
    "✓ Present";  
  
button.classList.remove(  
    "absent"  
);  
  
button.classList.add(  
    "present"  
);  
  
}  
  
}  
  
/* =====  
  
MARK ALL  
  
===== */  
  
function markAll(status) {  
  
    const buttons =
```

```
document.querySelectorAll(  
    "#attendanceTable .status"  
);
```

```
if (buttons.length === 0) {
```

```
    alert(  
        "Select subject and date first."  
    );
```

```
    return;
```

```
}
```

```
buttons.forEach(  
    button => {
```

```
        button.dataset.status =  
            status;
```

```
        if (  
            status === "Present"  
        ) {
```

```
button.textContent =
```

```
    "✓ Present";
```

```
button.classList.remove(
```

```
    "absent"
```

```
);
```

```
button.classList.add(
```

```
    "present"
```

```
);
```

```
} else {
```

```
button.textContent =
```

```
    "✗ Absent";
```

```
button.classList.remove(
```

```
    "present"
```

```
);
```

```
button.classList.add(
```

```
    "absent"
```

```
);
```

```
}
```

```

    }

);

}

/* =====

SAVE ATTENDANCE

===== */

function saveAttendance() {

    const subjectId =

        document.getElementById(

            "attendanceSubject"

        ).value;

    const date =

        document.getElementById(

            "attendanceDate"

        ).value;

    if (!subjectId) {

```

```
    alert(  
        "Please select a subject."  
    );  
  
    return;  
}
```

```
if (!date) {
```

```
    alert(  
        "Please select a date."  
    );  
  
    return;  
}
```

```
const buttons =  
    document.querySelectorAll(  
        "#attendanceTable .status"  
    );
```

```
if (buttons.length === 0) {
```

```
    alert(  
        "No students available."  
    );  
  
    return;  
}
```

```
buttons.forEach(  
    button => {  
  
        const studentId =  
            button.dataset.student;  
  
        const status =  
            button.dataset.status;  
  
        const existingIndex =  
            attendance.findIndex(  
                record =>  
  
                    String(  
                        record.studentId  
                    ) ===
```

String(studentId)

&&

String(

record.subjectId

) ===

String(subjectId)

&&

record.date ===

date

);

const record = {

id:

existingIndex >= 0

? attendance[

existingIndex

].id

: generateId(),

studentId:

studentId,

subjectId:

subjectId,

date:

date,

status:

status

};

if (

existingIndex >= 0

) {

attendance[

existingIndex

] = record;

} else {

attendance.push(

record

);

}

}

);

saveAttendance();

updateDashboard();

displayRecords();

displayPercentages();

alert(

"Attendance saved successfully! ✓"

);

}

/* =====

RECORDS

===== */

```
function displayRecords() {
```

```
    const table =
```

```
        document.getElementById(
```

```
            "recordsTable"
```

```
        );
```

```
    const subjectFilter =
```

```
        document.getElementById(
```

```
            "recordSubject"
```

```
        ).value;
```

```
    const dateFilter =
```

```
        document.getElementById(
```

```
            "recordDate"
```

```
        ).value;
```

```
    const search =
```

```
        document.getElementById(
```

```
            "recordSearch"
```

)?.value

.toLowerCase()

.trim() || "";

let records =

[...attendance];

if (subjectFilter) {

records =

records.filter(

record =>

String(

record.subjectId

) ===

String(subjectFilter)

);

}

if (dateFilter) {

records =

```
records.filter(  
  record =>  
    record.date ===  
    dateFilter  
);  
  
}
```

```
if (search) {
```

```
  records =
```

```
    records.filter(  
      record => {
```

```
        const student =
```

```
          students.find(  
            s =>
```

```
              String(  
                s.id
```

```
              ) ===
```

```
              String(  
                record.studentId
```

```
              )
```

```
            );
```

```
if (!student) {  
    return false;  
}
```

```
return (
```

```
    student.rollNo  
        .toLowerCase()  
        .includes(search)
```

```
||
```

```
    student.name  
        .toLowerCase()  
        .includes(search)
```

```
);
```

```
}
```

```
);
```

```
}
```

```
records.sort(  
    (a, b) =>  
        b.date.localeCompare(  
            a.date  
        )  
    );
```

```
table.innerHTML = "";
```

```
if (  
    records.length === 0  
) {
```

```
    table.innerHTML = `
```

```
        <tr>
```

```
            <td colspan="6"
```

```
                style="text-align:center">
```

```
                No attendance records found.
```

```
            </td>
```

</tr>

`;
`;

return;

}

records.forEach(
record => {

const student =

students.find(
s =>

String(s.id) ===

String(
record.studentId

)

);

const subject =

subjects.find(
s =>

String(s.id) ===

String(
subjectId

)

record.subjectId

)

);

if (

!student ||

!subject

) {

return;

}

const present =

record.status ===

"Present";

table.innerHTML += `

<tr>

<td>

`\${record.date}

</td>

<td>

`${escapeHTML(
student.rollNo
)}`

</td>

<td>

`${escapeHTML(
student.name
)}`

</td>

<td>

`${escapeHTML(
subject.code
)}`

-

`${escapeHTML(
subject.name
)}`

</td>

<td>

```
<span  
  class="status  
  ${present  
    ? "present"  
    : "absent"}">
```

```
  ${present  
    ? "✓ Present"  
    : "✗ Absent"}
```

```
</span>
```

```
</td>
```

```
<td>
```

```
<button  
  class="btn danger"  
  onclick="deleteAttendance('${record.id}')">
```

 Delete

```
</button>
```

```
</td>
```

```
</tr>
```

```
`;  
`;
```

```
}
```

```
);
```

```
}
```

```
/* =====
```

```
DELETE ATTENDANCE
```

```
===== */
```

```
function deleteAttendance(id) {
```

```
    if (
```

```
        !confirm(
```

```
            "Delete this attendance record?"
```

```
        )
```

```
    ){
```

```
        return;
```

```
    }
```

```
attendance =  
    attendance.filter(  
        record =>  
            String(record.id) !==  
            String(id)  
    );
```

```
saveAttendance();
```

```
refreshEverything();
```

```
alert(  
    "Attendance record deleted."  
);
```

```
}
```

```
/* =====  
  
ATTENDANCE PERCENTAGE  
  
===== */
```

```
function displayPercentages() {
```

```
    const table =
```

```
        document.getElementById(
```

```
            "percentageTable"
```

```
        );
```

```
    table.innerHTML = "";
```

```
    if (
```

```
        students.length === 0
```

```
    ) {
```

```
        table.innerHTML = `
```

```
            <tr>
```

```
                <td colspan="6"
```

```
                    style="text-align:center">
```

```
                        No students available.
```

```
            </td>
```

</tr>

`;
`;

return;

}

students.forEach(
student => {

const records =
attendance.filter(
record =>
String(
record.studentId
) ===
String(student.id)
);

const total =
records.length;

const present =

```
records.filter(  
  record =>  
    record.status ===  
      "Present"  
)
```

```
const absent =  
  records.filter(  
    record =>  
      record.status ===  
        "Absent"  
  )
```

```
const percentage =  
  total > 0
```

```
    ? (  
      present /  
      total *  
      100  
    )
```

```
    : 0;
```

```
let percentageClass =
```

```
    "good";
```

```
if (
```

```
    percentage < 75 &&
```

```
    percentage >= 60
```

```
) {
```

```
    percentageClass =
```

```
        "warning";
```

```
}
```

```
if (
```

```
    percentage < 60
```

```
) {
```

```
    percentageClass =
```

```
        "low";
```

```
}
```

```
table.innerHTML += `
```

```
<tr>
```

```
<td>
```

```
<strong>
```

```
    ${escapeHTML(  
        student.rollNo  
    )}
```

```
</strong>
```

```
</td>
```

```
<td>
```

```
    ${escapeHTML(  
        student.name  
    )}
```

```
</td>
```

```
<td>
```

```
    ${total}
```

```
</td>
```

```
<td>
```

```
    ${present}
```

```
</td>
```

<td>

 \${absent}

</td>

<td>

 <span

 class="percentage

 \${percentageClass}">

 \${percentage.toFixed(2)}%

</td>

</tr>

`;

}

);

}

```
/* =====
```

```
CLEAR ALL DATA
```

```
===== */
```

```
function clearAllData() {
```

```
    const confirmed =
```

```
        confirm(
```

```
            "WARNING!\n\n" +
```

```
            "This will permanently delete " +
```

```
            "all students, subjects and attendance " +
```

```
            "from this browser.\n\n" +
```

```
            "Continue?"
```

```
        );
```

```
    if (!confirmed) {
```

```
        return;
```

```
    }
```

```
    students = [];
```

```
    subjects = [];
```

```
    attendance = [];
```

```
localStorage.removeItem(  
    STUDENTS_KEY  
);
```

```
localStorage.removeItem(  
    SUBJECTS_KEY  
);
```

```
localStorage.removeItem(  
    ATTENDANCE_KEY  
);
```

```
refreshEverything();
```

```
alert(  
    "All data has been cleared."  
);
```

```
}
```

```
/* =====
```

HTML SECURITY

```
===== */
```

```
function escapeHTML(value) {
```

```
    return String(value)
```

```
        .replaceAll(
```

```
            "&",
```

```
            "&amp;"
```

```
        )
```

```
        .replaceAll(
```

```
            "<",
```

```
            "&lt;"
```

```
        )
```

```
        .replaceAll(
```

```
            ">",
```

```
            "&gt;"
```

```
        )
```

```
        .replaceAll(
```

```
            "'",
```

```
            "&quot;"
```

```
        )
```

```
.replaceAll(  
    "",  
    "&#039;"  
);  
  
}
```

Database.sql:-

```
CREATE DATABASE college_attendance;
```

```
USE college_attendance;
```

```
CREATE TABLE students (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    roll_no VARCHAR(30) NOT NULL UNIQUE,  
    name VARCHAR(100) NOT NULL,  
    department VARCHAR(100) NOT NULL,  
    semester INT NOT NULL,  
    division VARCHAR(10),  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

```
CREATE TABLE subjects (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    subject_code VARCHAR(30) NOT NULL UNIQUE,
```

```
subject_name VARCHAR(100) NOT NULL,  
  
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
  
);
```

```
CREATE TABLE attendance (  
  
    id INT AUTO_INCREMENT PRIMARY KEY,  
  
    student_id INT NOT NULL,  
  
    subject_id INT NOT NULL,  
  
    attendance_date DATE NOT NULL,  
  
    status ENUM('Present','Absent') NOT NULL,  
  
  
    FOREIGN KEY (student_id)  
        REFERENCES students(id)  
        ON DELETE CASCADE,  
  
  
    FOREIGN KEY (subject_id)  
        REFERENCES subjects(id)  
        ON DELETE CASCADE,  
  
  
    UNIQUE(student_id, subject_id, attendance_date)  
  
);
```

```
INSERT INTO students  
  
(roll_no, name, department, semester, division)  
  
VALUES  
  
('IT001', 'Rahul Patil', 'Information Technology', 5, 'B'),
```

('IT002', 'Priya Sharma', 'Information Technology', 5, 'B'),

('IT003', 'Amit Joshi', 'Information Technology', 5, 'B');

INSERT INTO subjects

(subject_code, subject_name)

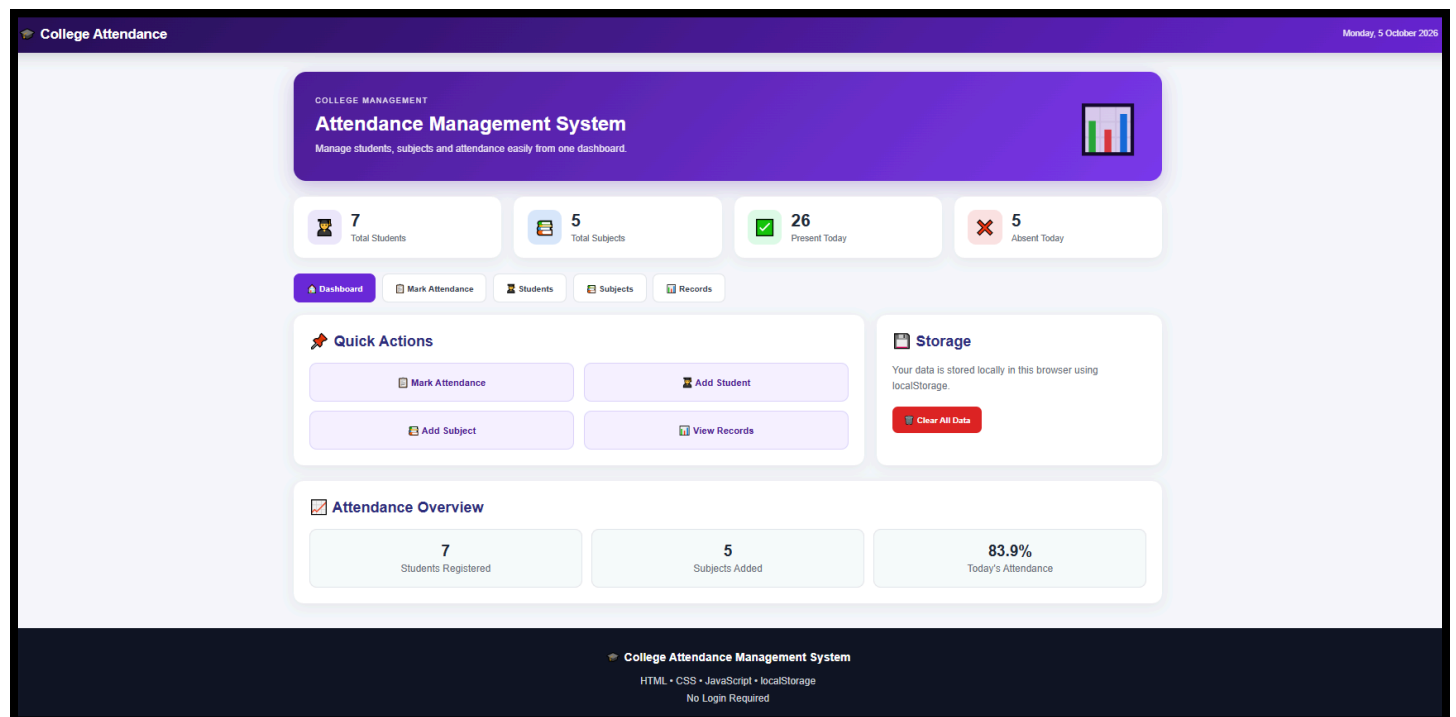
VALUES

('CS501', 'Database Management System'),

('CS502', 'Computer Networks'),

('CS503', 'Software Engineering');

OUTPUT:-



College Attendance

Monday, 5 October 2026

COLLEGE MANAGEMENT

Attendance Management System

Manage students, subjects and attendance easily from one dashboard.

7

Total Students

5

Total Subjects

26

Present Today

5

Absent Today

Dashboard

Mark Attendance

Students

Subjects

Records

Mark Attendance

Select subject and date.

Subject

CS058 - AI

Date

18/05/2026

✓ Mark All Present

✗ Mark All Absent

Roll No	Student Name	Department	Semester	Division	Status
IT201	Sugat bramhane	information technology	5	b	✓ Present
IT202	Abhilaash sharma	information technology	5	b	✓ Present
IT204	janvhi patil	information technology	5	b	✓ Present
IT205	priya deshmukh	information technology	5	b	✗ Absent
IT203	Riya chavan	information technology	5	b	✓ Present
IT206	sameer sharma	information technology	5	b	✓ Present
IT208	Nisha Chaturvedi	information technology	5	b	✓ Present

College Attendance

Monday, 5 October 2026

COLLEGE MANAGEMENT

Attendance Management System

Manage students, subjects and attendance easily from one dashboard.

7

Total Students

5

Total Subjects

26

Present Today

5

Absent Today

Dashboard

Mark Attendance

Students

Subjects

Records

Student Management

Roll Number*

CS0001

Student Name*

Enter student name

Department*

Computer Engineering

Semester*

5

Division

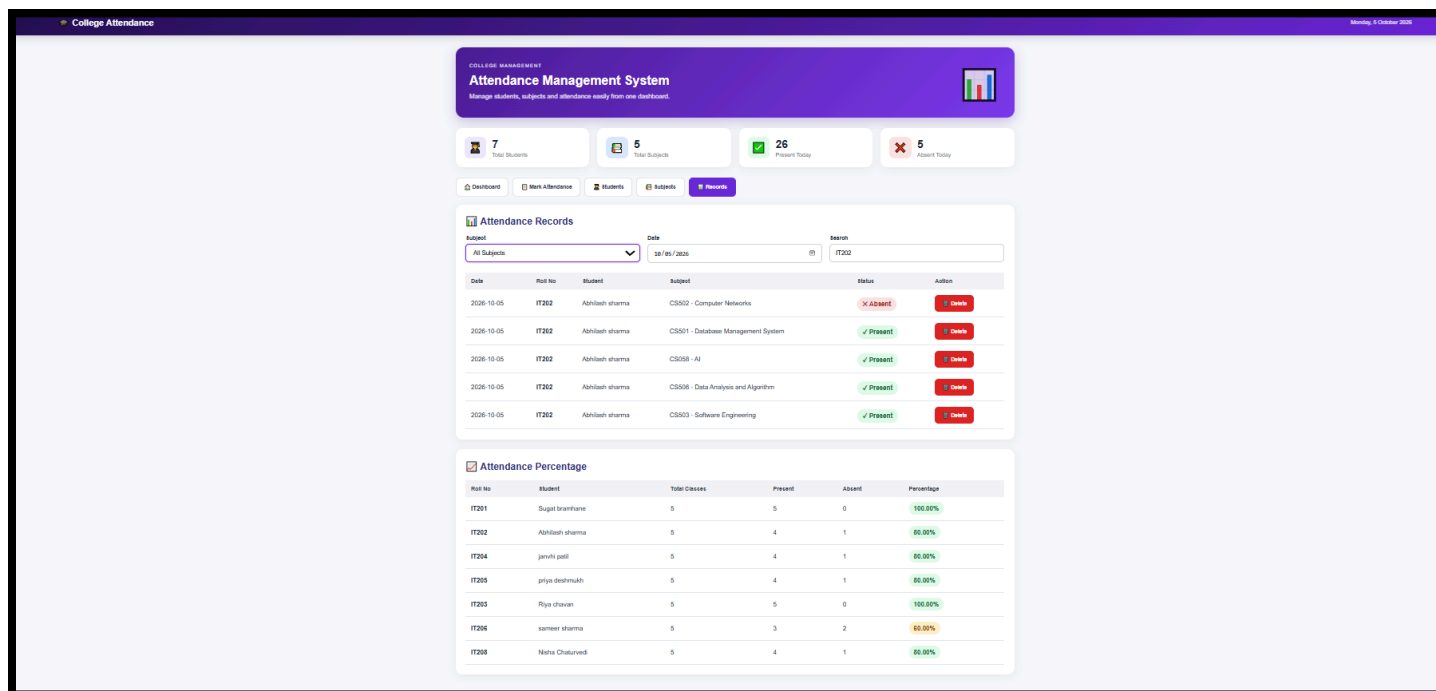
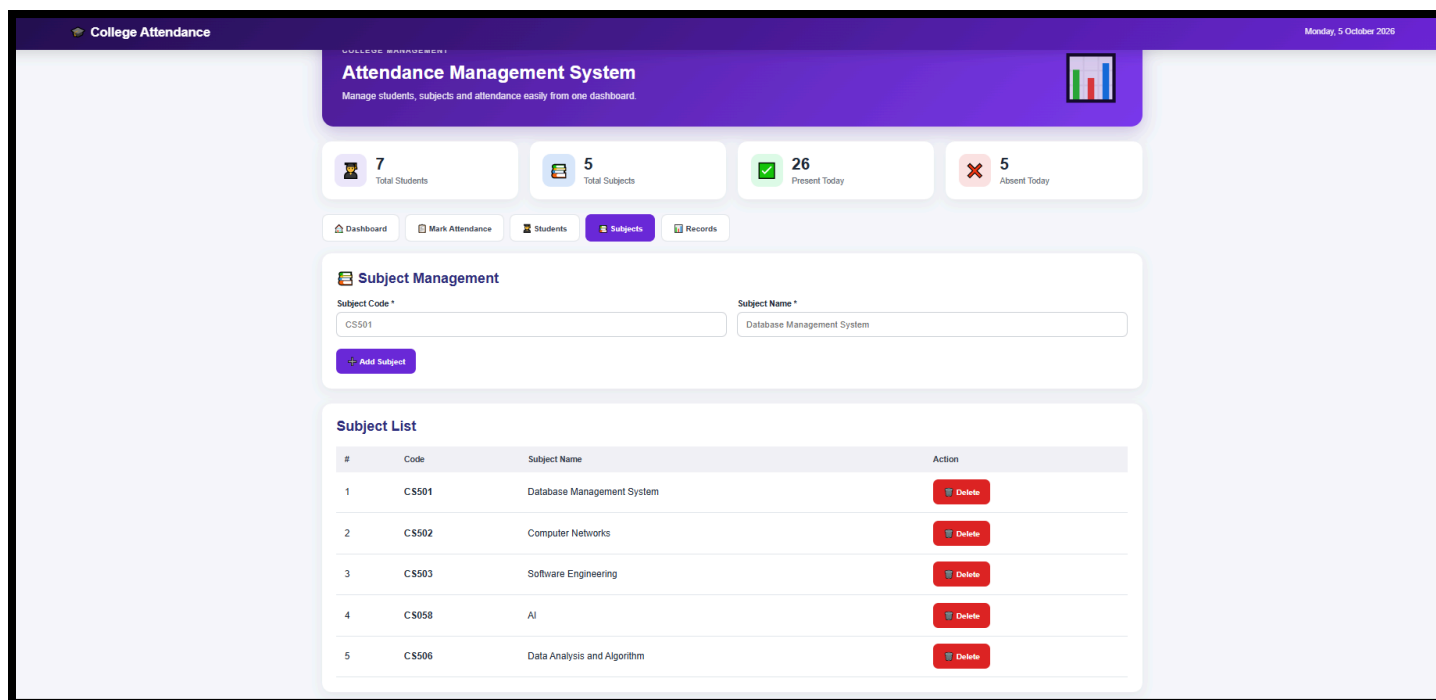
A

Add Student

Student List

#	Roll No	Name	Department	Semester	Division	Action
1	IT201	Sugat bramhane	information technology	5	b	Edit Delete
2	IT202	Abhilaash sharma	information technology	5	b	Edit Delete
3	IT204	janvhi patil	information technology	5	b	Edit Delete
4	IT205	priya deshmukh	information technology	5	b	Edit Delete
5	IT203	Riya chavan	information technology	5	b	Edit Delete
6	IT206	sameer sharma	information technology	5	b	Edit Delete
7	IT208	Nisha Chaturvedi	information technology	5	b	Edit Delete

College Attendance Management System



GITHUB LINK:-